PG No. 45 $C_{6\mathrm{h}}(c)$ $6/m$ [hexagonal] (lg basis)

bra: $= \langle s|$
ket: $= |s\rangle$

Table 1: (s,s) block.

No.	multipole	matrix
1	symmetry	1
	$\mathbb{Q}_0^{(a)}(A_g)$	$\begin{bmatrix} 1 \end{bmatrix}$

bra: $= \langle s|$
ket: $= |p_x\rangle, |p_y\rangle, |p_z\rangle$

Table 2: (s,p) block.

No.	multipole	matrix
2	symmetry	z
	$\mathbb{Q}_1^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$
3	symmetry	$\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$
	$\mathbb{Q}_1^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} \frac{1}{2} & \frac{i}{2} & 0 \end{bmatrix}$
4	symmetry	$\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$
	$\mathbb{Q}_1^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} \frac{1}{2} & -\frac{i}{2} & 0 \end{bmatrix}$
5	symmetry	z
	$\mathbb{T}_1^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{2} \end{bmatrix}$
6	symmetry	$\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$
	$\mathbb{T}_1^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} \frac{i}{2} & -\frac{1}{2} & 0 \end{bmatrix}$
7	symmetry	$\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$
	$\mathbb{T}_1^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} \frac{i}{2} & \frac{1}{2} & 0 \end{bmatrix}$

bra: $= \langle s|$
ket: $= |d_v\rangle, |d_{xy}\rangle, |d_{xz}\rangle, |d_{yz}\rangle, |d_u\rangle$

Table 3: (s,d) block.

No.	multipole	matrix
8	symmetry	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
	$\mathbb{Q}_2^{(a)}(A_g)$	$\begin{bmatrix}0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2}\end{bmatrix}$
9	symmetry	$\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$
	$\mathbb{Q}_2^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix}0 & 0 & \frac{1}{2} & \frac{i}{2} & 0\end{bmatrix}$
10	symmetry	$\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$
	$\mathbb{Q}_2^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix}0 & 0 & \frac{1}{2} & -\frac{i}{2} & 0\end{bmatrix}$
11	symmetry	$\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$
	$\mathbb{Q}_2^{(a)}(E_{2g}^{(a)})$	$\begin{bmatrix}\frac{1}{2} & -\frac{i}{2} & 0 & 0 & 0\end{bmatrix}$
12	symmetry	$\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$
	$\mathbb{Q}_2^{(a)}(E_{2g}^{(b)})$	$\begin{bmatrix}\frac{1}{2} & \frac{i}{2} & 0 & 0 & 0\end{bmatrix}$
13	symmetry	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
	$\mathbb{T}_2^{(a)}(A_g)$	$\begin{bmatrix}0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2}\end{bmatrix}$
14	symmetry	$\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$
	$\mathbb{T}_2^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix}0 & 0 & \frac{i}{2} & -\frac{1}{2} & 0\end{bmatrix}$
15	symmetry	$\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$
	$\mathbb{T}_2^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix}0 & 0 & \frac{i}{2} & \frac{1}{2} & 0\end{bmatrix}$
16	symmetry	$\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$
	$\mathbb{T}_2^{(a)}(E_{2g}^{(a)})$	$\begin{bmatrix}\frac{i}{2} & \frac{1}{2} & 0 & 0 & 0\end{bmatrix}$
17	symmetry	$\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$
	$\mathbb{T}_2^{(a)}(E_{2g}^{(b)})$	$\begin{bmatrix}\frac{i}{2} & -\frac{1}{2} & 0 & 0 & 0\end{bmatrix}$

bra: $= \langle s|$

ket: $= |f_c\rangle, |f_s\rangle, |f_z^\beta\rangle, |f_{xyz}\rangle, |f_x\rangle, |f_y\rangle, |f_z^\alpha\rangle$

Table 4: (s,f) block.

No.	multipole	matrix
18	symmetry	$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$

continued ...

Table 4

No.	multipole	matrix
	$\mathbb{Q}_3^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$
19	symmetry	$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$
	$\mathbb{Q}_3^{(a)}(B_u,1)$	$\begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
20	symmetry	$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$
	$\mathbb{Q}_3^{(a)}(B_u,2)$	$\begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
21	symmetry	$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$
	$\mathbb{Q}_3^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{i}{2} & 0 \end{bmatrix}$
22	symmetry	$-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$
	$\mathbb{Q}_3^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & -\frac{i}{2} & 0 \end{bmatrix}$
23	symmetry	$\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$
	$\mathbb{Q}_3^{(a)}(E_{2u}^{(a)})$	$\begin{bmatrix} 0 & 0 & \frac{1}{2} & -\frac{i}{2} & 0 & 0 & 0 \end{bmatrix}$
24	symmetry	$\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$
	$\mathbb{Q}_3^{(a)}(E_{2u}^{(b)})$	$\begin{bmatrix} 0 & 0 & \frac{1}{2} & \frac{i}{2} & 0 & 0 & 0 \end{bmatrix}$
25	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{T}_3^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \end{bmatrix}$
26	symmetry	$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$
	$\mathbb{T}_3^{(a)}(B_u,1)$	$\begin{bmatrix} 0 & \frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
27	symmetry	$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$
	$\mathbb{T}_3^{(a)}(B_u,2)$	$\begin{bmatrix} \frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
28	symmetry	$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$
	$\mathbb{T}_3^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & -\frac{1}{2} & 0 \end{bmatrix}$
29	symmetry	$-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$
	$\mathbb{T}_3^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & \frac{1}{2} & 0 \end{bmatrix}$
30	symmetry	$\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$
	$\mathbb{T}_3^{(a)}(E_{2u}^{(a)})$	$\begin{bmatrix} 0 & 0 & \frac{i}{2} & \frac{1}{2} & 0 & 0 & 0 \end{bmatrix}$
31	symmetry	$\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$
	$\mathbb{T}_3^{(a)}(E_{2u}^{(b)})$	$\begin{bmatrix} 0 & 0 & \frac{i}{2} & -\frac{1}{2} & 0 & 0 & 0 \end{bmatrix}$

bra: $= \langle p_x| , \langle p_y| , \langle p_z|$
ket: $= |p_x\rangle , |p_y\rangle , |p_z\rangle$

Table 5: (p,p) block.

No.	multipole	matrix
32	symmetry	1
	$\mathbb{Q}_0^{(a)}(A_g)$	$\begin{bmatrix} \frac{\sqrt{3}}{3} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$
33	symmetry	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
	$\mathbb{Q}_2^{(a)}(A_g)$	$\begin{bmatrix} -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{3} \end{bmatrix}$
34	symmetry	$\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$
	$\mathbb{Q}_2^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{i}{2} \\ \frac{1}{2} & \frac{i}{2} & 0 \end{bmatrix}$
35	symmetry	$\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$
	$\mathbb{Q}_2^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & \frac{1}{2} \\ 0 & 0 & -\frac{i}{2} \\ \frac{1}{2} & -\frac{i}{2} & 0 \end{bmatrix}$
36	symmetry	$\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{Q}_2^{(a)}(E_{2g}^{(a)})$	$\begin{bmatrix} \frac{1}{2} & -\frac{i}{2} & 0 \\ -\frac{i}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$
37	symmetry	$\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{Q}_2^{(a)}(E_{2g}^{(b)})$	$\begin{bmatrix} \frac{1}{2} & \frac{i}{2} & 0 \\ \frac{i}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$
38	symmetry	z

continued ...

Table 5

No.	multipole	matrix
	$\mathbb{M}_1^{(a)}(A_g)$	$\begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} & 0 \\ \frac{\sqrt{2}i}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
39	symmetry	$\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$
	$\mathbb{M}_1^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & -\frac{1}{2} \\ 0 & 0 & -\frac{i}{2} \\ \frac{1}{2} & \frac{i}{2} & 0 \end{bmatrix}$
40	symmetry	$\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$
	$\mathbb{M}_1^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & \frac{1}{2} \\ 0 & 0 & -\frac{i}{2} \\ -\frac{1}{2} & \frac{i}{2} & 0 \end{bmatrix}$

bra: $= \langle p_x| , \langle p_y| , \langle p_z|$
ket: $= |d_v\rangle , |d_{xy}\rangle , |d_{xz}\rangle , |d_{yz}\rangle , |d_u\rangle$

Table 6: (p,d) block.

No.	multipole	matrix
41	symmetry	z
	$\mathbb{Q}_1^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} \end{bmatrix}$
42	symmetry	$\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$
	$\mathbb{Q}_1^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} \frac{\sqrt{30}}{20} & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{10}}{20} \\ -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{10}i}{20} \\ 0 & 0 & \frac{\sqrt{30}}{20} & \frac{\sqrt{30}i}{20} & 0 \end{bmatrix}$
43	symmetry	$\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$

continued ...

Table 6

No.	multipole	matrix
	$\mathbb{Q}_1^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} \frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{10}}{20} \\ \frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{10}i}{20} \\ 0 & 0 & \frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{20} & 0 \end{bmatrix}$
44	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{Q}_3^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{10} \end{bmatrix}$
45	symmetry	$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$
	$\mathbb{Q}_3^{(a)}(B_u, 1)$	$\begin{bmatrix} 0 & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
46	symmetry	$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$
	$\mathbb{Q}_3^{(a)}(B_u, 2)$	$\begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
47	symmetry	$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$
	$\mathbb{Q}_3^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} -\frac{\sqrt{30}}{60} & -\frac{\sqrt{30}i}{60} & 0 & 0 & \frac{\sqrt{10}}{10} \\ \frac{\sqrt{30}i}{60} & -\frac{\sqrt{30}}{60} & 0 & 0 & \frac{\sqrt{10}i}{10} \\ 0 & 0 & \frac{\sqrt{30}}{15} & \frac{\sqrt{30}i}{15} & 0 \end{bmatrix}$
48	symmetry	$-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$
	$\mathbb{Q}_3^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} -\frac{\sqrt{30}}{60} & \frac{\sqrt{30}i}{60} & 0 & 0 & \frac{\sqrt{10}}{10} \\ -\frac{\sqrt{30}i}{60} & -\frac{\sqrt{30}}{60} & 0 & 0 & -\frac{\sqrt{10}i}{10} \\ 0 & 0 & \frac{\sqrt{30}}{15} & -\frac{\sqrt{30}i}{15} & 0 \end{bmatrix}$
49	symmetry	$\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$
	$\mathbb{Q}_3^{(a)}(E_{2u}^{(a)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$
50	symmetry	$\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$

continued ...

Table 6

No.	multipole	matrix
	$\mathbb{Q}_3^{(a)}(E_{2u}^{(b)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$
51	symmetry	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
	$\mathbb{G}_2^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
52	symmetry	$\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$
	$\mathbb{G}_2^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} -\frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ -\frac{\sqrt{6}}{12} & -\frac{\sqrt{6}i}{12} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & \frac{\sqrt{6}i}{12} & -\frac{\sqrt{6}}{12} & 0 \end{bmatrix}$
53	symmetry	$\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$
	$\mathbb{G}_2^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 & 0 & \frac{\sqrt{2}i}{4} \\ -\frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{12} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & -\frac{\sqrt{6}i}{12} & -\frac{\sqrt{6}}{12} & 0 \end{bmatrix}$
54	symmetry	$\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{G}_2^{(a)}(E_{2u}^{(a)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{12} & -\frac{\sqrt{6}i}{12} & 0 \\ -\frac{\sqrt{6}i}{6} & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 \end{bmatrix}$
55	symmetry	$\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{G}_2^{(a)}(E_{2u}^{(b)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{12} & 0 \\ \frac{\sqrt{6}i}{6} & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 \end{bmatrix}$
56	symmetry	z
	$\mathbb{T}_1^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} \end{bmatrix}$
57	symmetry	$\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$

continued ...

Table 6

No.	multipole	matrix
	$\mathbb{T}_1^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} \frac{\sqrt{30}i}{20} & -\frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{10}i}{20} \\ \frac{\sqrt{30}}{20} & \frac{\sqrt{30}i}{20} & 0 & 0 & \frac{\sqrt{10}}{20} \\ 0 & 0 & \frac{\sqrt{30}i}{20} & -\frac{\sqrt{30}}{20} & 0 \end{bmatrix}$
58	symmetry	$\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$
	$\mathbb{T}_1^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} \frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{10}i}{20} \\ -\frac{\sqrt{30}}{20} & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{10}}{20} \\ 0 & 0 & \frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 \end{bmatrix}$
59	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{T}_3^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{10} \end{bmatrix}$
60	symmetry	$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$
	$\mathbb{T}_3^{(a)}(B_u, 1)$	$\begin{bmatrix} 0 & \frac{i}{2} & 0 & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
61	symmetry	$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$
	$\mathbb{T}_3^{(a)}(B_u, 2)$	$\begin{bmatrix} \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
62	symmetry	$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$
	$\mathbb{T}_3^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} -\frac{\sqrt{30}i}{60} & \frac{\sqrt{30}}{60} & 0 & 0 & \frac{\sqrt{10}i}{10} \\ -\frac{\sqrt{30}}{60} & -\frac{\sqrt{30}i}{60} & 0 & 0 & -\frac{\sqrt{10}}{10} \\ 0 & 0 & \frac{\sqrt{30}i}{15} & -\frac{\sqrt{30}}{15} & 0 \end{bmatrix}$
63	symmetry	$-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$
	$\mathbb{T}_3^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} -\frac{\sqrt{30}i}{60} & -\frac{\sqrt{30}}{60} & 0 & 0 & \frac{\sqrt{10}i}{10} \\ \frac{\sqrt{30}}{60} & -\frac{\sqrt{30}i}{60} & 0 & 0 & \frac{\sqrt{10}}{10} \\ 0 & 0 & \frac{\sqrt{30}i}{15} & \frac{\sqrt{30}}{15} & 0 \end{bmatrix}$
64	symmetry	$\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$

continued ...

Table 6

No.	multipole	matrix
	$\mathbb{T}_3^{(a)}(E_{2u}^{(a)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}i}{6} & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}i}{6} & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$
65	symmetry	$\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$
	$\mathbb{T}_3^{(a)}(E_{2u}^{(b)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$
66	symmetry	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
	$\mathbb{M}_2^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
67	symmetry	$\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$
	$\mathbb{M}_2^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} -\frac{\sqrt{6}}{12} & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{2}}{4} \\ \frac{\sqrt{6}i}{12} & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{12} & 0 \end{bmatrix}$
68	symmetry	$\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$
	$\mathbb{M}_2^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} \frac{\sqrt{6}}{12} & -\frac{\sqrt{6}i}{12} & 0 & 0 & \frac{\sqrt{2}}{4} \\ \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & -\frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{12} & 0 \end{bmatrix}$
69	symmetry	$\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{M}_2^{(a)}(E_{2u}^{(a)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}}{12} & -\frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{12} & -\frac{\sqrt{6}}{12} & 0 \\ -\frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \end{bmatrix}$
70	symmetry	$\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{M}_2^{(a)}(E_{2u}^{(b)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{12} & -\frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \end{bmatrix}$

bra: $= \langle p_x|, \langle p_y|, \langle p_z|$

ket: $= |f_c\rangle, |f_s\rangle, |f_z^\beta\rangle, |f_{xyz}\rangle, |f_x\rangle, |f_y\rangle, |f_z^\alpha\rangle$

Table 7: (p,f) block.

No.	multipole	matrix
71	symmetry	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
	$\mathbb{Q}_2^{(a)}(A_g)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} \end{bmatrix}$
72	symmetry	$\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$
	$\mathbb{Q}_2^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{105}}{42} & \frac{\sqrt{105}i}{42} & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & 0 & -\frac{\sqrt{105}i}{42} & \frac{\sqrt{105}}{42} & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{21} & \frac{\sqrt{42}i}{21} & 0 \end{bmatrix}$
73	symmetry	$\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$
	$\mathbb{Q}_2^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{105}}{42} & -\frac{\sqrt{105}i}{42} & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & 0 & \frac{\sqrt{105}i}{42} & \frac{\sqrt{105}}{42} & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{21} & -\frac{\sqrt{42}i}{21} & 0 \end{bmatrix}$
74	symmetry	$\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{Q}_2^{(a)}(E_{2g}^{(a)})$	$\begin{bmatrix} \frac{\sqrt{70}}{28} & -\frac{\sqrt{70}i}{28} & 0 & 0 & -\frac{\sqrt{42}}{84} & \frac{\sqrt{42}i}{84} & 0 \\ \frac{\sqrt{70}i}{28} & \frac{\sqrt{70}}{28} & 0 & 0 & \frac{\sqrt{42}i}{84} & \frac{\sqrt{42}}{84} & 0 \\ 0 & 0 & \frac{\sqrt{105}}{42} & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 \end{bmatrix}$
75	symmetry	$\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{Q}_2^{(a)}(E_{2g}^{(b)})$	$\begin{bmatrix} \frac{\sqrt{70}}{28} & \frac{\sqrt{70}i}{28} & 0 & 0 & -\frac{\sqrt{42}}{84} & -\frac{\sqrt{42}i}{84} & 0 \\ -\frac{\sqrt{70}i}{28} & \frac{\sqrt{70}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{84} & \frac{\sqrt{42}}{84} & 0 \\ 0 & 0 & \frac{\sqrt{105}}{42} & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 \end{bmatrix}$
76	symmetry	$\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$
	$\mathbb{Q}_4^{(a)}(A_g)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{7} \end{bmatrix}$
77	symmetry	$\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$

continued ...

Table 7

No.	multipole	matrix
	$\mathbb{Q}_4^{(a)}(B_g,1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{4} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
78	symmetry	$\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$
	$\mathbb{Q}_4^{(a)}(B_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{3}}{4} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
79	symmetry	$-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$
	$\mathbb{Q}_4^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}}{56} & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{70}}{28} \\ 0 & 0 & \frac{\sqrt{42}i}{56} & -\frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{70}i}{28} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{28} & \frac{\sqrt{105}i}{28} & 0 \end{bmatrix}$
80	symmetry	$-\frac{3\sqrt{5}x^3z}{4} + \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 + \frac{3\sqrt{5}iy^3z}{4} - \sqrt{5}iyz^3$
	$\mathbb{Q}_4^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}}{56} & \frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{70}}{28} \\ 0 & 0 & -\frac{\sqrt{42}i}{56} & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{70}i}{28} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{28} & -\frac{\sqrt{105}i}{28} & 0 \end{bmatrix}$
81	symmetry	$\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$
	$\mathbb{Q}_4^{(a)}(E_{2g}^{(a)},1)$	$\begin{bmatrix} \frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
82	symmetry	$-\frac{\sqrt{10}x^4}{8} + \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} + \frac{\sqrt{10}ixy^3}{4} - \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$
	$\mathbb{Q}_4^{(a)}(E_{2g}^{(a)},2)$	$\begin{bmatrix} -\frac{\sqrt{14}}{56} & \frac{\sqrt{14}i}{56} & 0 & 0 & \frac{\sqrt{210}}{56} & -\frac{\sqrt{210}i}{56} & 0 \\ -\frac{\sqrt{14}i}{56} & -\frac{\sqrt{14}}{56} & 0 & 0 & -\frac{\sqrt{210}i}{56} & -\frac{\sqrt{210}}{56} & 0 \\ 0 & 0 & \frac{\sqrt{21}}{14} & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 \end{bmatrix}$
83	symmetry	$\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$
	$\mathbb{Q}_4^{(a)}(E_{2g}^{(b)},1)$	$\begin{bmatrix} \frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
84	symmetry	$-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$

continued ...

Table 7

No.	multipole	matrix
	$\mathbb{Q}_4^{(a)}(E_{2g}^{(b)},2)$	$\begin{bmatrix} -\frac{\sqrt{14}}{56} & -\frac{\sqrt{14}i}{56} & 0 & 0 & \frac{\sqrt{210}}{56} & \frac{\sqrt{210}i}{56} & 0 \\ \frac{\sqrt{14}i}{56} & -\frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{210}i}{56} & -\frac{\sqrt{210}}{56} & 0 \\ 0 & 0 & \frac{\sqrt{21}}{14} & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 \end{bmatrix}$
85	symmetry	$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$
	$\mathbb{G}_3^{(a)}(A_g)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
86	symmetry	$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$
	$\mathbb{G}_3^{(a)}(B_g,1)$	$\begin{bmatrix} 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
87	symmetry	$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$
	$\mathbb{G}_3^{(a)}(B_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{4} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
88	symmetry	$-\frac{\sqrt{3}x^3}{4}-\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2-\frac{\sqrt{3}iy^3}{4}+\sqrt{3}iyz^2$
	$\mathbb{G}_3^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{30}i}{24} & \frac{\sqrt{30}}{24} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & -\frac{\sqrt{30}}{24} & -\frac{\sqrt{30}i}{24} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & -\frac{\sqrt{3}}{12} & 0 \end{bmatrix}$
89	symmetry	$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$
	$\mathbb{G}_3^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}i}{24} & \frac{\sqrt{30}}{24} & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & -\frac{\sqrt{30}}{24} & \frac{\sqrt{30}i}{24} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & -\frac{\sqrt{3}}{12} & 0 \end{bmatrix}$
90	symmetry	$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$
	$\mathbb{G}_3^{(a)}(E_{2g}^{(a)})$	$\begin{bmatrix} \frac{\sqrt{2}i}{8} & \frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{30}i}{24} & \frac{\sqrt{30}}{24} & 0 \\ -\frac{\sqrt{2}}{8} & \frac{\sqrt{2}i}{8} & 0 & 0 & \frac{\sqrt{30}}{24} & -\frac{\sqrt{30}i}{24} & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$
91	symmetry	$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$

continued ...

Table 7

No.	multipole	matrix
	$\mathbb{G}_3^{(a)}(E_{2g}^{(b)})$	$\begin{bmatrix} -\frac{\sqrt{2}i}{8} & \frac{\sqrt{2}}{8} & 0 & 0 & -\frac{\sqrt{30}i}{24} & \frac{\sqrt{30}}{24} & 0 \\ -\frac{\sqrt{2}}{8} & -\frac{\sqrt{2}i}{8} & 0 & 0 & \frac{\sqrt{30}}{24} & \frac{\sqrt{30}i}{24} & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$
92	symmetry	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
	$\mathbb{T}_2^{(a)}(A_g)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} \end{bmatrix}$
93	symmetry	$\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$
	$\mathbb{T}_2^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{105}i}{42} & -\frac{\sqrt{105}}{42} & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & 0 & \frac{\sqrt{105}}{42} & \frac{\sqrt{105}i}{42} & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & -\frac{\sqrt{42}}{21} & 0 \end{bmatrix}$
94	symmetry	$\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$
	$\mathbb{T}_2^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{105}i}{42} & \frac{\sqrt{105}}{42} & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & 0 & -\frac{\sqrt{105}}{42} & \frac{\sqrt{105}i}{42} & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & \frac{\sqrt{42}}{21} & 0 \end{bmatrix}$
95	symmetry	$\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{T}_2^{(a)}(E_{2g}^{(a)})$	$\begin{bmatrix} \frac{\sqrt{70}i}{28} & \frac{\sqrt{70}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{84} & -\frac{\sqrt{42}}{84} & 0 \\ -\frac{\sqrt{70}}{28} & \frac{\sqrt{70}i}{28} & 0 & 0 & -\frac{\sqrt{42}}{84} & \frac{\sqrt{42}i}{84} & 0 \\ 0 & 0 & \frac{\sqrt{105}i}{42} & \frac{\sqrt{105}}{42} & 0 & 0 & 0 \end{bmatrix}$
96	symmetry	$\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{T}_2^{(a)}(E_{2g}^{(b)})$	$\begin{bmatrix} \frac{\sqrt{70}i}{28} & -\frac{\sqrt{70}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{84} & \frac{\sqrt{42}}{84} & 0 \\ \frac{\sqrt{70}}{28} & \frac{\sqrt{70}i}{28} & 0 & 0 & \frac{\sqrt{42}}{84} & \frac{\sqrt{42}i}{84} & 0 \\ 0 & 0 & \frac{\sqrt{105}i}{42} & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 \end{bmatrix}$
97	symmetry	$\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$
	$\mathbb{T}_4^{(a)}(A_g)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{7} \end{bmatrix}$
98	symmetry	$\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$

continued ...

Table 7

No.	multipole	matrix
	$\mathbb{T}_4^{(a)}(B_g,1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{4} & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
99	symmetry	$\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$
	$\mathbb{T}_4^{(a)}(B_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{3}i}{4} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
100	symmetry	$-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$
	$\mathbb{T}_4^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}i}{56} & \frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{70}i}{28} \\ 0 & 0 & -\frac{\sqrt{42}}{56} & -\frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{70}}{28} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & -\frac{\sqrt{105}}{28} & 0 \end{bmatrix}$
101	symmetry	$-\frac{3\sqrt{5}x^3z}{4} + \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 + \frac{3\sqrt{5}iy^3z}{4} - \sqrt{5}iyz^3$
	$\mathbb{T}_4^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}i}{56} & -\frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{70}i}{28} \\ 0 & 0 & \frac{\sqrt{42}}{56} & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{70}}{28} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & \frac{\sqrt{105}}{28} & 0 \end{bmatrix}$
102	symmetry	$\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$
	$\mathbb{T}_4^{(a)}(E_{2g}^{(a)},1)$	$\begin{bmatrix} \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
103	symmetry	$-\frac{\sqrt{10}x^4}{8} + \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} + \frac{\sqrt{10}ixy^3}{4} - \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$
	$\mathbb{T}_4^{(a)}(E_{2g}^{(a)},2)$	$\begin{bmatrix} -\frac{\sqrt{14}i}{56} & -\frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{210}i}{56} & \frac{\sqrt{210}}{56} & 0 \\ \frac{\sqrt{14}}{56} & -\frac{\sqrt{14}i}{56} & 0 & 0 & \frac{\sqrt{210}}{56} & -\frac{\sqrt{210}i}{56} & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{14} & \frac{\sqrt{21}}{14} & 0 & 0 & 0 \end{bmatrix}$
104	symmetry	$\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$
	$\mathbb{T}_4^{(a)}(E_{2g}^{(b)},1)$	$\begin{bmatrix} \frac{\sqrt{2}i}{4} & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
105	symmetry	$-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$

continued ...

Table 7

No.	multipole	matrix
	$\mathbb{T}_4^{(a)}(E_{2g}^{(b)},2)$	$\begin{bmatrix} -\frac{\sqrt{14}i}{56} & \frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{210}i}{56} & -\frac{\sqrt{210}}{56} & 0 \\ -\frac{\sqrt{14}}{56} & -\frac{\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{210}}{56} & -\frac{\sqrt{210}i}{56} & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{14} & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 \end{bmatrix}$
106	symmetry	$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$
	$\mathbb{M}_3^{(a)}(A_g)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
107	symmetry	$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$
	$\mathbb{M}_3^{(a)}(B_g,1)$	$\begin{bmatrix} 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
108	symmetry	$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$
	$\mathbb{M}_3^{(a)}(B_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 \\ 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{4} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
109	symmetry	$-\frac{\sqrt{3}x^3}{4}-\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2-\frac{\sqrt{3}iy^3}{4}+\sqrt{3}iyz^2$
	$\mathbb{M}_3^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{30}}{24} & -\frac{\sqrt{30}i}{24} & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & \frac{\sqrt{30}i}{24} & -\frac{\sqrt{30}}{24} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & \frac{\sqrt{3}i}{12} & 0 \end{bmatrix}$
110	symmetry	$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$
	$\mathbb{M}_3^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}}{24} & -\frac{\sqrt{30}i}{24} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & \frac{\sqrt{30}i}{24} & \frac{\sqrt{30}}{24} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & \frac{\sqrt{3}i}{12} & 0 \end{bmatrix}$
111	symmetry	$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$
	$\mathbb{M}_3^{(a)}(E_{2g}^{(a)})$	$\begin{bmatrix} \frac{\sqrt{2}}{8} & -\frac{\sqrt{2}i}{8} & 0 & 0 & \frac{\sqrt{30}}{24} & -\frac{\sqrt{30}i}{24} & 0 \\ \frac{\sqrt{2}i}{8} & \frac{\sqrt{2}}{8} & 0 & 0 & -\frac{\sqrt{30}i}{24} & -\frac{\sqrt{30}}{24} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$
112	symmetry	$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$

continued ...

Table 7

No.	multipole	matrix
	$\mathbb{M}_3^{(a)}(E_{2g}^{(b)})$	$\begin{bmatrix} -\frac{\sqrt{2}}{8} & -\frac{\sqrt{2}i}{8} & 0 & 0 & -\frac{\sqrt{30}}{24} & -\frac{\sqrt{30}i}{24} & 0 \\ \frac{\sqrt{2}i}{8} & -\frac{\sqrt{2}}{8} & 0 & 0 & -\frac{\sqrt{30}i}{24} & \frac{\sqrt{30}}{24} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$

bra: $= \langle d_v|, \langle d_{xy}|, \langle d_{xz}|, \langle d_{yz}|, \langle d_u|$
ket: $= |d_v\rangle, |d_{xy}\rangle, |d_{xz}\rangle, |d_{yz}\rangle, |d_u\rangle$

Table 8: (d,d) block.

No.	multipole	matrix
113	symmetry	1
	$\mathbb{Q}_0^{(a)}(A_g)$	$\begin{bmatrix} \frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{5} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{5} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{5} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} \end{bmatrix}$
114	symmetry	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
	$\mathbb{Q}_2^{(a)}(A_g)$	$\begin{bmatrix} -\frac{\sqrt{14}}{7} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{7} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{7} \end{bmatrix}$
115	symmetry	$\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$
	$\mathbb{Q}_2^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{21}}{14} & -\frac{\sqrt{21}i}{14} & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{14} & \frac{\sqrt{21}}{14} & 0 \\ \frac{\sqrt{21}}{14} & \frac{\sqrt{21}i}{14} & 0 & 0 & \frac{\sqrt{7}}{14} \\ -\frac{\sqrt{21}i}{14} & \frac{\sqrt{21}}{14} & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & 0 & \frac{\sqrt{7}}{14} & \frac{\sqrt{7}i}{14} & 0 \end{bmatrix}$
116	symmetry	$\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$

continued ...

Table 8

No.	multipole	matrix
	$\mathbb{Q}_2^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{21}}{14} & \frac{\sqrt{21}i}{14} & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{14} & \frac{\sqrt{21}}{14} & 0 \\ \frac{\sqrt{21}}{14} & -\frac{\sqrt{21}i}{14} & 0 & 0 & \frac{\sqrt{7}}{14} \\ \frac{\sqrt{21}i}{14} & \frac{\sqrt{21}}{14} & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & 0 & \frac{\sqrt{7}}{14} & -\frac{\sqrt{7}i}{14} & 0 \end{bmatrix}$
117	symmetry	$\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{Q}_2^{(a)}(E_{2g}^{(a)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{7} \\ 0 & 0 & \frac{\sqrt{21}}{14} & -\frac{\sqrt{21}i}{14} & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{14} & -\frac{\sqrt{21}}{14} & 0 \\ -\frac{\sqrt{7}}{7} & \frac{\sqrt{7}i}{7} & 0 & 0 & 0 \end{bmatrix}$
118	symmetry	$\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{Q}_2^{(a)}(E_{2g}^{(b)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{7} \\ 0 & 0 & \frac{\sqrt{21}}{14} & \frac{\sqrt{21}i}{14} & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{14} & -\frac{\sqrt{21}}{14} & 0 \\ -\frac{\sqrt{7}}{7} & -\frac{\sqrt{7}i}{7} & 0 & 0 & 0 \end{bmatrix}$
119	symmetry	$\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$
	$\mathbb{Q}_4^{(a)}(A_g)$	$\begin{bmatrix} \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{70}}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{70}}{35} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{35} \end{bmatrix}$
120	symmetry	$\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$

continued ...

Table 8

No.	multipole	matrix
	$\mathbb{Q}_4^{(a)}(B_g,1)$	$\begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
121	symmetry	$\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$
	$\mathbb{Q}_4^{(a)}(B_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
122	symmetry	$-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$
	$\mathbb{Q}_4^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{14}}{28} & \frac{\sqrt{14}i}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{14}i}{28} & -\frac{\sqrt{14}}{28} & 0 \\ -\frac{\sqrt{14}}{28} & -\frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{42}}{14} \\ \frac{\sqrt{14}i}{28} & -\frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{42}i}{14} \\ 0 & 0 & \frac{\sqrt{42}}{14} & \frac{\sqrt{42}i}{14} & 0 \end{bmatrix}$
123	symmetry	$-\frac{3\sqrt{5}x^3z}{4} + \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 + \frac{3\sqrt{5}iy^3z}{4} - \sqrt{5}iyz^3$
	$\mathbb{Q}_4^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{14}}{28} & -\frac{\sqrt{14}i}{28} & 0 \\ 0 & 0 & \frac{\sqrt{14}i}{28} & -\frac{\sqrt{14}}{28} & 0 \\ -\frac{\sqrt{14}}{28} & \frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{42}}{14} \\ -\frac{\sqrt{14}i}{28} & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{42}i}{14} \\ 0 & 0 & \frac{\sqrt{42}}{14} & -\frac{\sqrt{42}i}{14} & 0 \end{bmatrix}$
124	symmetry	$\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$

continued ...

Table 8

No.	multipole	matrix
	$\mathbb{Q}_4^{(a)}(E_{2g}^{(a)},1)$	$\begin{bmatrix} \frac{1}{2} & \frac{i}{2} & 0 & 0 & 0 \\ \frac{i}{2} & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
125	symmetry	$-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$
	$\mathbb{Q}_4^{(a)}(E_{2g}^{(a)},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} \\ 0 & 0 & \frac{\sqrt{7}}{7} & -\frac{\sqrt{7}i}{7} & 0 \\ 0 & 0 & -\frac{\sqrt{7}i}{7} & -\frac{\sqrt{7}}{7} & 0 \\ \frac{\sqrt{21}}{14} & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 \end{bmatrix}$
126	symmetry	$\frac{\sqrt{70}x^4}{16}-\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}+\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$
	$\mathbb{Q}_4^{(a)}(E_{2g}^{(b)},1)$	$\begin{bmatrix} \frac{1}{2} & -\frac{i}{2} & 0 & 0 & 0 \\ -\frac{i}{2} & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
127	symmetry	$-\frac{\sqrt{10}x^4}{8}-\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}-\frac{\sqrt{10}ixy^3}{4}+\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$
	$\mathbb{Q}_4^{(a)}(E_{2g}^{(b)},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} \\ 0 & 0 & \frac{\sqrt{7}}{7} & \frac{\sqrt{7}i}{7} & 0 \\ 0 & 0 & \frac{\sqrt{7}i}{7} & -\frac{\sqrt{7}}{7} & 0 \\ \frac{\sqrt{21}}{14} & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 \end{bmatrix}$
128	symmetry	z

continued ...

Table 8

No.	multipole	matrix
	$\mathbb{M}_1^{(a)}(A_g)$	$\begin{bmatrix} 0 & -\frac{\sqrt{10}i}{5} & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
129	symmetry	$\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$
	$\mathbb{M}_1^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{5}}{10} & \frac{\sqrt{5}i}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{10} & -\frac{\sqrt{5}}{10} & 0 \\ \frac{\sqrt{5}}{10} & \frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{15}}{10} \\ -\frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{15}i}{10} \\ 0 & 0 & \frac{\sqrt{15}}{10} & \frac{\sqrt{15}i}{10} & 0 \end{bmatrix}$
130	symmetry	$\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$
	$\mathbb{M}_1^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{5}}{10} & \frac{\sqrt{5}i}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{10} & 0 \\ -\frac{\sqrt{5}}{10} & \frac{\sqrt{5}i}{10} & 0 & 0 & \frac{\sqrt{15}}{10} \\ -\frac{\sqrt{5}i}{10} & -\frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{15}i}{10} \\ 0 & 0 & -\frac{\sqrt{15}}{10} & \frac{\sqrt{15}i}{10} & 0 \end{bmatrix}$
131	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{M}_3^{(a)}(A_g)$	$\begin{bmatrix} 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{5} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{5} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
132	symmetry	$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$

continued ...

Table 8

No.	multipole	matrix
	$\mathbb{M}_3^{(a)}(B_g,1)$	$\begin{bmatrix} 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
133	symmetry	$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$
	$\mathbb{M}_3^{(a)}(B_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & -\frac{i}{2} & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
134	symmetry	$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$
	$\mathbb{M}_3^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{20} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 \\ -\frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{10}}{10} \\ \frac{\sqrt{30}i}{20} & -\frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{10}i}{10} \\ 0 & 0 & \frac{\sqrt{10}}{10} & \frac{\sqrt{10}i}{10} & 0 \end{bmatrix}$
135	symmetry	$-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$
	$\mathbb{M}_3^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{20} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{20} & -\frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{20} & 0 & 0 & \frac{\sqrt{10}}{10} \\ \frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{10}i}{10} \\ 0 & 0 & -\frac{\sqrt{10}}{10} & \frac{\sqrt{10}i}{10} & 0 \end{bmatrix}$
136	symmetry	$\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$

continued ...

Table 8

No.	multipole	matrix
	$\mathbb{M}_3^{(a)}(E_{2g}^{(a)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{2} & \frac{i}{2} & 0 & 0 & 0 \end{bmatrix}$
137	symmetry	$\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$
	$\mathbb{M}_3^{(a)}(E_{2g}^{(b)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{i}{2} & 0 & 0 & 0 \end{bmatrix}$

bra: $= \langle d_v| , \langle d_{xy}| , \langle d_{xz}| , \langle d_{yz}| , \langle d_u|$
ket: $= |f_c\rangle , |f_s\rangle , |f_z^\beta\rangle , |f_{xyz}\rangle , |f_x\rangle , |f_y\rangle , |f_z^\alpha\rangle$

Table 9: (d,f) block.

No.	multipole	matrix
138	symmetry	z
	$\mathbb{Q}_1^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} \end{bmatrix}$
139	symmetry	$\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{Q}_1^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} \frac{\sqrt{42}}{28} & \frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{\sqrt{70}}{140} & \frac{\sqrt{70}i}{140} & 0 \\ -\frac{\sqrt{42}i}{28} & \frac{\sqrt{42}}{28} & 0 & 0 & -\frac{\sqrt{70}i}{140} & -\frac{\sqrt{70}}{140} & 0 \\ 0 & 0 & \frac{\sqrt{7}}{14} & \frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{105}}{70} \\ 0 & 0 & -\frac{\sqrt{7}i}{14} & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{105}i}{70} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{70} & \frac{\sqrt{210}i}{70} & 0 \end{bmatrix}$
140	symmetry	$\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$
	$\mathbb{Q}_1^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} \frac{\sqrt{42}}{28} & -\frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{\sqrt{70}}{140} & -\frac{\sqrt{70}i}{140} & 0 \\ \frac{\sqrt{42}i}{28} & \frac{\sqrt{42}}{28} & 0 & 0 & \frac{\sqrt{70}i}{140} & -\frac{\sqrt{70}}{140} & 0 \\ 0 & 0 & \frac{\sqrt{7}}{14} & -\frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{105}}{70} \\ 0 & 0 & \frac{\sqrt{7}i}{14} & \frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{105}i}{70} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{70} & -\frac{\sqrt{210}i}{70} & 0 \end{bmatrix}$
141	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{Q}_3^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{15} \end{bmatrix}$
142	symmetry	$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$
	$\mathbb{Q}_3^{(a)}(B_u, 1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{12} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
143	symmetry	$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{Q}_3^{(a)}(B_u,2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{12} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
144	symmetry	$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$
	$\mathbb{Q}_3^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} -\frac{\sqrt{3}}{12} & -\frac{\sqrt{3}i}{12} & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \\ \frac{\sqrt{3}i}{12} & -\frac{\sqrt{3}}{12} & 0 & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{8} & \frac{\sqrt{2}i}{8} & 0 & 0 & \frac{\sqrt{30}}{60} \\ 0 & 0 & -\frac{\sqrt{2}i}{8} & \frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{30}i}{60} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & \frac{\sqrt{15}i}{20} & 0 \end{bmatrix}$
145	symmetry	$-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$
	$\mathbb{Q}_3^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} -\frac{\sqrt{3}}{12} & \frac{\sqrt{3}i}{12} & 0 & 0 & \frac{\sqrt{5}}{10} & \frac{\sqrt{5}i}{10} & 0 \\ -\frac{\sqrt{3}i}{12} & -\frac{\sqrt{3}}{12} & 0 & 0 & -\frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{8} & -\frac{\sqrt{2}i}{8} & 0 & 0 & \frac{\sqrt{30}}{60} \\ 0 & 0 & \frac{\sqrt{2}i}{8} & \frac{\sqrt{2}}{8} & 0 & 0 & -\frac{\sqrt{30}i}{60} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & -\frac{\sqrt{15}i}{20} & 0 \end{bmatrix}$
146	symmetry	$\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$
	$\mathbb{Q}_3^{(a)}(E_{2u}^{(a)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \\ \frac{\sqrt{30}}{24} & -\frac{\sqrt{30}i}{24} & 0 & 0 & \frac{\sqrt{2}}{8} & -\frac{\sqrt{2}i}{8} & 0 \\ \frac{\sqrt{30}i}{24} & \frac{\sqrt{30}}{24} & 0 & 0 & -\frac{\sqrt{2}i}{8} & -\frac{\sqrt{2}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
147	symmetry	$\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{Q}_3^{(a)}(E_{2u}^{(b)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ \frac{\sqrt{30}}{24} & \frac{\sqrt{30}i}{24} & 0 & 0 & \frac{\sqrt{2}}{8} & \frac{\sqrt{2}i}{8} & 0 \\ -\frac{\sqrt{30}i}{24} & \frac{\sqrt{30}}{24} & 0 & 0 & \frac{\sqrt{2}i}{8} & -\frac{\sqrt{2}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
148	symmetry	$\frac{15x^4z}{8} + \frac{15x^2y^2z}{4} - 5x^2z^3 + \frac{15y^4z}{8} - 5y^2z^3 + z^5$
	$\mathbb{Q}_5^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{21} \end{bmatrix}$
149	symmetry	$-\frac{3\sqrt{70}x^4y}{16} - \frac{\sqrt{70}x^2y^3}{8} + \frac{3\sqrt{70}x^2yz^2}{2} + \frac{\sqrt{70}y^5}{16} - \frac{\sqrt{70}y^3z^2}{2}$
	$\mathbb{Q}_5^{(a)}(B_u, 1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
150	symmetry	$-\frac{\sqrt{70}x^5}{16} + \frac{\sqrt{70}x^3y^2}{8} + \frac{\sqrt{70}x^3z^2}{2} + \frac{3\sqrt{70}xy^4}{16} - \frac{3\sqrt{70}xy^2z^2}{2}$
	$\mathbb{Q}_5^{(a)}(B_u, 2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{15} & 0 & 0 & 0 \\ \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
151	symmetry	$\frac{3\sqrt{7}x^5}{16} - \frac{15\sqrt{7}ix^4y}{16} - \frac{15\sqrt{7}x^3y^2}{8} + \frac{15\sqrt{7}ix^2y^3}{8} + \frac{15\sqrt{7}xy^4}{16} - \frac{3\sqrt{7}iy^5}{16}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{Q}_5^{(a)}(E_{1u}^{(a)},1)$	$\begin{bmatrix} \frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
152	symmetry	$\frac{\sqrt{30}x^5}{16}+\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}+\frac{\sqrt{30}ix^2y^3}{8}-\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}+\frac{\sqrt{30}iy^5}{16}-\frac{3\sqrt{30}iy^3z^2}{4}+\frac{\sqrt{30}iyz^4}{2}$
	$\mathbb{Q}_5^{(a)}(E_{1u}^{(a)},2)$	$\begin{bmatrix} \frac{\sqrt{105}}{420} & \frac{\sqrt{105}i}{420} & 0 & 0 & -\frac{\sqrt{7}}{28} & \frac{\sqrt{7}i}{28} & 0 \\ -\frac{\sqrt{105}i}{420} & \frac{\sqrt{105}}{420} & 0 & 0 & -\frac{\sqrt{7}i}{28} & -\frac{\sqrt{7}}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{70} & -\frac{\sqrt{70}i}{70} & 0 & 0 & \frac{\sqrt{42}}{21} \\ 0 & 0 & \frac{\sqrt{70}i}{70} & -\frac{\sqrt{70}}{70} & 0 & 0 & \frac{\sqrt{42}i}{21} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & \frac{\sqrt{21}i}{14} & 0 \end{bmatrix}$
153	symmetry	$\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$
	$\mathbb{Q}_5^{(a)}(E_{1u}^{(b)},1)$	$\begin{bmatrix} \frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
154	symmetry	$\frac{\sqrt{30}x^5}{16}-\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}-\frac{\sqrt{30}ix^2y^3}{8}+\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}-\frac{\sqrt{30}iy^5}{16}+\frac{3\sqrt{30}iy^3z^2}{4}-\frac{\sqrt{30}iyz^4}{2}$
	$\mathbb{Q}_5^{(a)}(E_{1u}^{(b)},2)$	$\begin{bmatrix} \frac{\sqrt{105}}{420} & -\frac{\sqrt{105}i}{420} & 0 & 0 & -\frac{\sqrt{7}}{28} & -\frac{\sqrt{7}i}{28} & 0 \\ \frac{\sqrt{105}i}{420} & \frac{\sqrt{105}}{420} & 0 & 0 & \frac{\sqrt{7}i}{28} & -\frac{\sqrt{7}}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{70} & \frac{\sqrt{70}i}{70} & 0 & 0 & \frac{\sqrt{42}}{21} \\ 0 & 0 & -\frac{\sqrt{70}i}{70} & -\frac{\sqrt{70}}{70} & 0 & 0 & -\frac{\sqrt{42}i}{21} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & -\frac{\sqrt{21}i}{14} & 0 \end{bmatrix}$
155	symmetry	$\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{Q}_5^{(a)}(E_{2u}^{(a)},1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}}{20} & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{20} & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ \frac{\sqrt{5}}{10} & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{10} & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
156	symmetry	$-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$
	$\mathbb{Q}_5^{(a)}(E_{2u}^{(a)},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ -\frac{\sqrt{15}}{60} & \frac{\sqrt{15}i}{60} & 0 & 0 & \frac{1}{4} & -\frac{i}{4} & 0 \\ -\frac{\sqrt{15}i}{60} & -\frac{\sqrt{15}}{60} & 0 & 0 & -\frac{i}{4} & -\frac{1}{4} & 0 \\ 0 & 0 & \frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 \end{bmatrix}$
157	symmetry	$\frac{3\sqrt{70}x^4z}{16}-\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}+\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$
	$\mathbb{Q}_5^{(a)}(E_{2u}^{(b)},1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{20} & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
158	symmetry	$-\frac{\sqrt{210}x^4z}{8}-\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}-\frac{\sqrt{210}ixy^3z}{4}+\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$
	$\mathbb{Q}_5^{(a)}(E_{2u}^{(b)},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ -\frac{\sqrt{15}}{60} & -\frac{\sqrt{15}i}{60} & 0 & 0 & \frac{1}{4} & \frac{i}{4} & 0 \\ \frac{\sqrt{15}i}{60} & -\frac{\sqrt{15}}{60} & 0 & 0 & \frac{i}{4} & -\frac{1}{4} & 0 \\ 0 & 0 & \frac{\sqrt{30}}{20} & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 \end{bmatrix}$
159	symmetry	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{G}_2^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{35}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
160	symmetry	$\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$
	$\mathbb{G}_2^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} -\frac{\sqrt{35}i}{28} & \frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{21}i}{28} & -\frac{\sqrt{21}}{28} & 0 \\ -\frac{\sqrt{35}}{28} & -\frac{\sqrt{35}i}{28} & 0 & 0 & \frac{\sqrt{21}}{28} & -\frac{\sqrt{21}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & -\frac{\sqrt{7}}{14} & 0 \end{bmatrix}$
161	symmetry	$\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$
	$\mathbb{G}_2^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} \frac{\sqrt{35}i}{28} & \frac{\sqrt{35}}{28} & 0 & 0 & \frac{\sqrt{21}i}{28} & -\frac{\sqrt{21}}{28} & 0 \\ -\frac{\sqrt{35}}{28} & \frac{\sqrt{35}i}{28} & 0 & 0 & \frac{\sqrt{21}}{28} & \frac{\sqrt{21}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & -\frac{\sqrt{7}}{14} & 0 \end{bmatrix}$
162	symmetry	$\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{G}_2^{(a)}(E_{2u}^{(a)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} \\ -\frac{\sqrt{35}i}{28} & -\frac{\sqrt{35}}{28} & 0 & 0 & \frac{\sqrt{21}i}{28} & \frac{\sqrt{21}}{28} & 0 \\ \frac{\sqrt{35}}{28} & -\frac{\sqrt{35}i}{28} & 0 & 0 & \frac{\sqrt{21}}{28} & -\frac{\sqrt{21}i}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{28} & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 \end{bmatrix}$
163	symmetry	$\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{G}_2^{(a)}(E_{2u}^{(b)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} \\ \frac{\sqrt{35}i}{28} & -\frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{21}i}{28} & \frac{\sqrt{21}}{28} & 0 \\ \frac{\sqrt{35}}{28} & \frac{\sqrt{35}i}{28} & 0 & 0 & \frac{\sqrt{21}}{28} & \frac{\sqrt{21}i}{28} & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{28} & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 \end{bmatrix}$
164	symmetry	$\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$
	$\mathbb{G}_4^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{14} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
165	symmetry	$\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$
	$\mathbb{G}_4^{(a)}(B_u, 1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
166	symmetry	$\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$
	$\mathbb{G}_4^{(a)}(B_u, 2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 \\ \frac{3\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
167	symmetry	$-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{G}_4^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} \frac{\sqrt{105}i}{140} & -\frac{\sqrt{105}}{140} & 0 & 0 & \frac{\sqrt{7}i}{14} & \frac{\sqrt{7}}{14} & 0 \\ \frac{\sqrt{105}}{140} & \frac{\sqrt{105}i}{140} & 0 & 0 & -\frac{\sqrt{7}}{14} & \frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{40} & \frac{\sqrt{70}}{40} & 0 & 0 & -\frac{\sqrt{42}i}{28} \\ 0 & 0 & -\frac{\sqrt{70}}{40} & -\frac{\sqrt{70}i}{40} & 0 & 0 & \frac{\sqrt{42}}{28} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & -\frac{\sqrt{21}}{28} & 0 \end{bmatrix}$
168	symmetry	$-\frac{3\sqrt{5}x^3z}{4} + \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 + \frac{3\sqrt{5}iy^3z}{4} - \sqrt{5}iyz^3$
	$\mathbb{G}_4^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} -\frac{\sqrt{105}i}{140} & -\frac{\sqrt{105}}{140} & 0 & 0 & -\frac{\sqrt{7}i}{14} & \frac{\sqrt{7}}{14} & 0 \\ \frac{\sqrt{105}}{140} & -\frac{\sqrt{105}i}{140} & 0 & 0 & -\frac{\sqrt{7}}{14} & -\frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{40} & \frac{\sqrt{70}}{40} & 0 & 0 & \frac{\sqrt{42}i}{28} \\ 0 & 0 & -\frac{\sqrt{70}}{40} & \frac{\sqrt{70}i}{40} & 0 & 0 & \frac{\sqrt{42}}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & -\frac{\sqrt{21}}{28} & 0 \end{bmatrix}$
169	symmetry	$\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$
	$\mathbb{G}_4^{(a)}(E_{2u}^{(a)},1)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{10} & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{20} & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
170	symmetry	$-\frac{\sqrt{10}x^4}{8} + \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} + \frac{\sqrt{10}ixy^3}{4} - \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$
	$\mathbb{G}_4^{(a)}(E_{2u}^{(a)},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} \\ \frac{3\sqrt{210}i}{280} & \frac{3\sqrt{210}}{280} & 0 & 0 & \frac{\sqrt{14}i}{56} & \frac{\sqrt{14}}{56} & 0 \\ -\frac{3\sqrt{210}}{280} & \frac{3\sqrt{210}i}{280} & 0 & 0 & \frac{\sqrt{14}}{56} & -\frac{\sqrt{14}i}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{35} & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 \end{bmatrix}$
171	symmetry	$\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{G}_4^{(a)}(E_{2u}^{(b)},1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{20} & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{20} & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
172	symmetry	$-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$
	$\mathbb{G}_4^{(a)}(E_{2u}^{(b)},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} \\ -\frac{3\sqrt{210}i}{280} & \frac{3\sqrt{210}}{280} & 0 & 0 & -\frac{\sqrt{14}i}{56} & \frac{\sqrt{14}}{56} & 0 \\ -\frac{3\sqrt{210}}{280} & -\frac{3\sqrt{210}i}{280} & 0 & 0 & \frac{\sqrt{14}}{56} & \frac{\sqrt{14}i}{56} & 0 \\ 0 & 0 & \frac{\sqrt{105}i}{35} & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 \end{bmatrix}$
173	symmetry	z
	$\mathbb{T}_1^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{35}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}i}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} \end{bmatrix}$
174	symmetry	$\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$
	$\mathbb{T}_1^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} \frac{\sqrt{42}i}{28} & -\frac{\sqrt{42}}{28} & 0 & 0 & -\frac{\sqrt{70}i}{140} & -\frac{\sqrt{70}}{140} & 0 \\ \frac{\sqrt{42}}{28} & \frac{\sqrt{42}i}{28} & 0 & 0 & \frac{\sqrt{70}}{140} & -\frac{\sqrt{70}i}{140} & 0 \\ 0 & 0 & \frac{\sqrt{7}i}{14} & -\frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{105}i}{70} \\ 0 & 0 & \frac{\sqrt{7}}{14} & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{105}}{70} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} & -\frac{\sqrt{210}}{70} & 0 \end{bmatrix}$
175	symmetry	$\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{T}_1^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} \frac{\sqrt{42}i}{28} & \frac{\sqrt{42}}{28} & 0 & 0 & -\frac{\sqrt{70}i}{140} & \frac{\sqrt{70}}{140} & 0 \\ -\frac{\sqrt{42}}{28} & \frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{\sqrt{70}}{140} & -\frac{\sqrt{70}i}{140} & 0 \\ 0 & 0 & \frac{\sqrt{7}i}{14} & \frac{\sqrt{7}}{14} & 0 & 0 & -\frac{\sqrt{105}i}{70} \\ 0 & 0 & -\frac{\sqrt{7}}{14} & \frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{105}}{70} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} & \frac{\sqrt{210}}{70} & 0 \end{bmatrix}$
176	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{T}_3^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{15} \end{bmatrix}$
177	symmetry	$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$
	$\mathbb{T}_3^{(a)}(B_u, 1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
178	symmetry	$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$
	$\mathbb{T}_3^{(a)}(B_u, 2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
179	symmetry	$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{T}_3^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} -\frac{\sqrt{3}i}{12} & \frac{\sqrt{3}}{12} & 0 & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{10} & 0 \\ -\frac{\sqrt{3}}{12} & -\frac{\sqrt{3}i}{12} & 0 & 0 & -\frac{\sqrt{5}}{10} & \frac{\sqrt{5}i}{10} & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{8} & -\frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{30}i}{60} \\ 0 & 0 & \frac{\sqrt{2}}{8} & \frac{\sqrt{2}i}{8} & 0 & 0 & -\frac{\sqrt{30}}{60} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & -\frac{\sqrt{15}}{20} & 0 \end{bmatrix}$
180	symmetry	$-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$
	$\mathbb{T}_3^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} -\frac{\sqrt{3}i}{12} & -\frac{\sqrt{3}}{12} & 0 & 0 & \frac{\sqrt{5}i}{10} & -\frac{\sqrt{5}}{10} & 0 \\ \frac{\sqrt{3}}{12} & -\frac{\sqrt{3}i}{12} & 0 & 0 & \frac{\sqrt{5}}{10} & \frac{\sqrt{5}i}{10} & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{8} & \frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{30}i}{60} \\ 0 & 0 & -\frac{\sqrt{2}}{8} & \frac{\sqrt{2}i}{8} & 0 & 0 & \frac{\sqrt{30}}{60} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & \frac{\sqrt{15}}{20} & 0 \end{bmatrix}$
181	symmetry	$\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$
	$\mathbb{T}_3^{(a)}(E_{2u}^{(a)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ \frac{\sqrt{30}i}{24} & \frac{\sqrt{30}}{24} & 0 & 0 & \frac{\sqrt{2}i}{8} & \frac{\sqrt{2}}{8} & 0 \\ -\frac{\sqrt{30}}{24} & \frac{\sqrt{30}i}{24} & 0 & 0 & \frac{\sqrt{2}}{8} & -\frac{\sqrt{2}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
182	symmetry	$\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$
	$\mathbb{T}_3^{(a)}(E_{2u}^{(b)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ \frac{\sqrt{30}i}{24} & -\frac{\sqrt{30}}{24} & 0 & 0 & \frac{\sqrt{2}i}{8} & -\frac{\sqrt{2}}{8} & 0 \\ \frac{\sqrt{30}}{24} & \frac{\sqrt{30}i}{24} & 0 & 0 & -\frac{\sqrt{2}}{8} & -\frac{\sqrt{2}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
183	symmetry	$\frac{15x^4z}{8} + \frac{15x^2y^2z}{4} - 5x^2z^3 + \frac{15y^4z}{8} - 5y^2z^3 + z^5$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{T}_5^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{21} \end{bmatrix}$
184	symmetry	$-\frac{3\sqrt{70}x^4y}{16} - \frac{\sqrt{70}x^2y^3}{8} + \frac{3\sqrt{70}x^2yz^2}{2} + \frac{\sqrt{70}y^5}{16} - \frac{\sqrt{70}y^3z^2}{2}$
	$\mathbb{T}_5^{(a)}(B_u,1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
185	symmetry	$-\frac{\sqrt{70}x^5}{16} + \frac{\sqrt{70}x^3y^2}{8} + \frac{\sqrt{70}x^3z^2}{2} + \frac{3\sqrt{70}xy^4}{16} - \frac{3\sqrt{70}xy^2z^2}{2}$
	$\mathbb{T}_5^{(a)}(B_u,2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{15} & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
186	symmetry	$\frac{3\sqrt{7}x^5}{16} - \frac{15\sqrt{7}ix^4y}{16} - \frac{15\sqrt{7}x^3y^2}{8} + \frac{15\sqrt{7}ix^2y^3}{8} + \frac{15\sqrt{7}xy^4}{16} - \frac{3\sqrt{7}iy^5}{16}$
	$\mathbb{T}_5^{(a)}(E_{1u}^{(a)},1)$	$\begin{bmatrix} \frac{\sqrt{2}i}{4} & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
187	symmetry	$\frac{\sqrt{30}x^5}{16} + \frac{\sqrt{30}ix^4y}{16} + \frac{\sqrt{30}x^3y^2}{8} - \frac{3\sqrt{30}x^3z^2}{4} + \frac{\sqrt{30}ix^2y^3}{8} - \frac{3\sqrt{30}ix^2yz^2}{4} + \frac{\sqrt{30}xy^4}{16} - \frac{3\sqrt{30}xy^2z^2}{4} + \frac{\sqrt{30}xz^4}{2} + \frac{\sqrt{30}iy^5}{16} - \frac{3\sqrt{30}iy^3z^2}{4} + \frac{\sqrt{30}iyz^4}{2}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{T}_5^{(a)}(E_{1u}^{(a)},2)$	$\begin{bmatrix} \frac{\sqrt{105}i}{420} & -\frac{\sqrt{105}}{420} & 0 & 0 & -\frac{\sqrt{7}i}{28} & -\frac{\sqrt{7}}{28} & 0 \\ \frac{\sqrt{105}}{420} & \frac{\sqrt{105}i}{420} & 0 & 0 & \frac{\sqrt{7}}{28} & -\frac{\sqrt{7}i}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{70} & \frac{\sqrt{70}}{70} & 0 & 0 & \frac{\sqrt{42}i}{21} \\ 0 & 0 & -\frac{\sqrt{70}}{70} & -\frac{\sqrt{70}i}{70} & 0 & 0 & -\frac{\sqrt{42}}{21} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & -\frac{\sqrt{21}}{14} & 0 \end{bmatrix}$
188	symmetry	$\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$
	$\mathbb{T}_5^{(a)}(E_{1u}^{(b)},1)$	$\begin{bmatrix} \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
189	symmetry	$\frac{\sqrt{30}x^5}{16}-\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}-\frac{\sqrt{30}ix^2y^3}{8}+\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}-\frac{\sqrt{30}iy^5}{16}+\frac{3\sqrt{30}iy^3z^2}{4}-\frac{\sqrt{30}iyz^4}{2}$
	$\mathbb{T}_5^{(a)}(E_{1u}^{(b)},2)$	$\begin{bmatrix} \frac{\sqrt{105}i}{420} & \frac{\sqrt{105}}{420} & 0 & 0 & -\frac{\sqrt{7}i}{28} & \frac{\sqrt{7}}{28} & 0 \\ -\frac{\sqrt{105}}{420} & \frac{\sqrt{105}i}{420} & 0 & 0 & -\frac{\sqrt{7}}{28} & -\frac{\sqrt{7}i}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{70} & -\frac{\sqrt{70}}{70} & 0 & 0 & \frac{\sqrt{42}i}{21} \\ 0 & 0 & \frac{\sqrt{70}}{70} & -\frac{\sqrt{70}i}{70} & 0 & 0 & \frac{\sqrt{42}}{21} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & \frac{\sqrt{21}}{14} & 0 \end{bmatrix}$
190	symmetry	$\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$
	$\mathbb{T}_5^{(a)}(E_{2u}^{(a)},1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}i}{20} & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{10} & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
191	symmetry	$-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{T}_5^{(a)}(E_{2u}^{(a)},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ -\frac{\sqrt{15}i}{60} & -\frac{\sqrt{15}}{60} & 0 & 0 & \frac{i}{4} & \frac{1}{4} & 0 \\ \frac{\sqrt{15}}{60} & -\frac{\sqrt{15}i}{60} & 0 & 0 & \frac{1}{4} & -\frac{i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 & 0 & 0 \end{bmatrix}$
192	symmetry	$\frac{3\sqrt{70}x^4z}{16} - \frac{3\sqrt{70}ix^3yz}{4} - \frac{9\sqrt{70}x^2y^2z}{8} + \frac{3\sqrt{70}ixy^3z}{4} + \frac{3\sqrt{70}y^4z}{16}$
	$\mathbb{T}_5^{(a)}(E_{2u}^{(b)},1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
193	symmetry	$-\frac{\sqrt{210}x^4z}{8} - \frac{\sqrt{210}ix^3yz}{4} + \frac{\sqrt{210}x^2z^3}{4} - \frac{\sqrt{210}ixy^3z}{4} + \frac{\sqrt{210}ixyz^3}{2} + \frac{\sqrt{210}y^4z}{8} - \frac{\sqrt{210}y^2z^3}{4}$
	$\mathbb{T}_5^{(a)}(E_{2u}^{(b)},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ -\frac{\sqrt{15}i}{60} & \frac{\sqrt{15}}{60} & 0 & 0 & \frac{i}{4} & -\frac{1}{4} & 0 \\ -\frac{\sqrt{15}}{60} & -\frac{\sqrt{15}i}{60} & 0 & 0 & -\frac{1}{4} & -\frac{i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{20} & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 \end{bmatrix}$
194	symmetry	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
	$\mathbb{M}_2^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{35}i}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
195	symmetry	$\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{M}_2^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} -\frac{\sqrt{35}}{28} & -\frac{\sqrt{35}i}{28} & 0 & 0 & -\frac{\sqrt{21}}{28} & \frac{\sqrt{21}i}{28} & 0 \\ \frac{\sqrt{35}i}{28} & -\frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{21}i}{28} & -\frac{\sqrt{21}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & \frac{\sqrt{7}i}{14} & 0 \end{bmatrix}$
196	symmetry	$\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$
	$\mathbb{M}_2^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} \frac{\sqrt{35}}{28} & -\frac{\sqrt{35}i}{28} & 0 & 0 & \frac{\sqrt{21}}{28} & \frac{\sqrt{21}i}{28} & 0 \\ \frac{\sqrt{35}i}{28} & \frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{21}i}{28} & \frac{\sqrt{21}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & \frac{\sqrt{7}i}{14} & 0 \end{bmatrix}$
197	symmetry	$\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{M}_2^{(a)}(E_{2u}^{(a)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} \\ -\frac{\sqrt{35}}{28} & \frac{\sqrt{35}i}{28} & 0 & 0 & \frac{\sqrt{21}}{28} & -\frac{\sqrt{21}i}{28} & 0 \\ -\frac{\sqrt{35}i}{28} & -\frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{21}i}{28} & -\frac{\sqrt{21}}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{28} & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 \end{bmatrix}$
198	symmetry	$\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{M}_2^{(a)}(E_{2u}^{(b)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} \\ \frac{\sqrt{35}}{28} & \frac{\sqrt{35}i}{28} & 0 & 0 & -\frac{\sqrt{21}}{28} & -\frac{\sqrt{21}i}{28} & 0 \\ -\frac{\sqrt{35}i}{28} & \frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{21}i}{28} & \frac{\sqrt{21}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{70}}{28} & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 \end{bmatrix}$
199	symmetry	$\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{M}_4^{(a)}(A_u)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{14} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
200	symmetry	$\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$
	$\mathbb{M}_4^{(a)}(B_u,1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
201	symmetry	$\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$
	$\mathbb{M}_4^{(a)}(B_u,2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 \\ -\frac{3\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
202	symmetry	$-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$
	$\mathbb{M}_4^{(a)}(E_{1u}^{(a)})$	$\begin{bmatrix} \frac{\sqrt{105}}{140} & \frac{\sqrt{105}i}{140} & 0 & 0 & \frac{\sqrt{7}}{14} & -\frac{\sqrt{7}i}{14} & 0 \\ -\frac{\sqrt{105}i}{140} & \frac{\sqrt{105}}{140} & 0 & 0 & \frac{\sqrt{7}i}{14} & \frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{40} & -\frac{\sqrt{70}i}{40} & 0 & 0 & -\frac{\sqrt{42}}{28} \\ 0 & 0 & \frac{\sqrt{70}i}{40} & -\frac{\sqrt{70}}{40} & 0 & 0 & -\frac{\sqrt{42}i}{28} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & \frac{\sqrt{21}i}{28} & 0 \end{bmatrix}$
203	symmetry	$-\frac{3\sqrt{5}x^3z}{4} + \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 + \frac{3\sqrt{5}iy^3z}{4} - \sqrt{5}iyz^3$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{M}_4^{(a)}(E_{1u}^{(b)})$	$\begin{bmatrix} -\frac{\sqrt{105}}{140} & \frac{\sqrt{105}i}{140} & 0 & 0 & -\frac{\sqrt{7}}{14} & -\frac{\sqrt{7}i}{14} & 0 \\ -\frac{\sqrt{105}i}{140} & -\frac{\sqrt{105}}{140} & 0 & 0 & \frac{\sqrt{7}i}{14} & -\frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & \frac{\sqrt{70}}{40} & -\frac{\sqrt{70}i}{40} & 0 & 0 & \frac{\sqrt{42}}{28} \\ 0 & 0 & \frac{\sqrt{70}i}{40} & \frac{\sqrt{70}}{40} & 0 & 0 & -\frac{\sqrt{42}i}{28} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & \frac{\sqrt{21}i}{28} & 0 \end{bmatrix}$
204	symmetry	$\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$
	$\mathbb{M}_4^{(a)}(E_{2u}^{(a)}, 1)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{20} & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{20} & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
205	symmetry	$-\frac{\sqrt{10}x^4}{8} + \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} + \frac{\sqrt{10}ixy^3}{4} - \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$
	$\mathbb{M}_4^{(a)}(E_{2u}^{(a)}, 2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} \\ \frac{3\sqrt{210}}{280} & -\frac{3\sqrt{210}i}{280} & 0 & 0 & \frac{\sqrt{14}}{56} & -\frac{\sqrt{14}i}{56} & 0 \\ \frac{3\sqrt{210}i}{280} & \frac{3\sqrt{210}}{280} & 0 & 0 & -\frac{\sqrt{14}i}{56} & -\frac{\sqrt{14}}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{35} & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 \end{bmatrix}$
206	symmetry	$\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$
	$\mathbb{M}_4^{(a)}(E_{2u}^{(b)}, 1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{10} & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{20} & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
207	symmetry	$-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{M}_4^{(a)}(E_{2u}^{(b)},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} \\ -\frac{3\sqrt{210}}{280} & -\frac{3\sqrt{210}i}{280} & 0 & 0 & -\frac{\sqrt{14}}{56} & -\frac{\sqrt{14}i}{56} & 0 \\ \frac{3\sqrt{210}i}{280} & -\frac{3\sqrt{210}}{280} & 0 & 0 & -\frac{\sqrt{14}i}{56} & \frac{\sqrt{14}}{56} & 0 \\ 0 & 0 & \frac{\sqrt{105}}{35} & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 \end{bmatrix}$

bra: $= \langle f_c|, \langle f_s|, \langle f_z^\beta|, \langle f_{xyz}|, \langle f_x|, \langle f_y|, \langle f_z^\alpha|$
ket: $= |f_c\rangle, |f_s\rangle, |f_z^\beta\rangle, |f_{xyz}\rangle, |f_x\rangle, |f_y\rangle, |f_z^\alpha\rangle$

Table 10: (f,f) block.

No.	multipole	matrix
208	symmetry	1
	$\mathbb{Q}_0^{(a)}(A_g)$	$\begin{bmatrix} \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} \end{bmatrix}$
209	symmetry	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
	$\mathbb{Q}_2^{(a)}(A_g)$	$\begin{bmatrix} -\frac{5\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{21} \end{bmatrix}$
210	symmetry	$\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_2^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & \frac{5\sqrt{21}}{84} & -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{21}i}{84} & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 \\ \frac{5\sqrt{21}}{84} & \frac{5\sqrt{21}i}{84} & 0 & 0 & \frac{\sqrt{35}}{28} & -\frac{\sqrt{35}i}{28} & 0 \\ -\frac{5\sqrt{21}i}{84} & \frac{5\sqrt{21}}{84} & 0 & 0 & \frac{\sqrt{35}i}{28} & \frac{\sqrt{35}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{35}}{28} & \frac{\sqrt{35}i}{28} & 0 & 0 & \frac{\sqrt{21}}{42} \\ 0 & 0 & -\frac{\sqrt{35}i}{28} & \frac{\sqrt{35}}{28} & 0 & 0 & \frac{\sqrt{21}i}{42} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & \frac{\sqrt{21}i}{42} & 0 \end{bmatrix}$
211	symmetry	$\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$
	$\mathbb{Q}_2^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & \frac{5\sqrt{21}}{84} & \frac{5\sqrt{21}i}{84} & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{21}i}{84} & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 \\ \frac{5\sqrt{21}}{84} & -\frac{5\sqrt{21}i}{84} & 0 & 0 & \frac{\sqrt{35}}{28} & \frac{\sqrt{35}i}{28} & 0 \\ \frac{5\sqrt{21}i}{84} & \frac{5\sqrt{21}}{84} & 0 & 0 & -\frac{\sqrt{35}i}{28} & \frac{\sqrt{35}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{35}}{28} & -\frac{\sqrt{35}i}{28} & 0 & 0 & \frac{\sqrt{21}}{42} \\ 0 & 0 & \frac{\sqrt{35}i}{28} & \frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{21}i}{42} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & -\frac{\sqrt{21}i}{42} & 0 \end{bmatrix}$
212	symmetry	$\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$
	$\mathbb{Q}_2^{(a)}(E_{2g}^{(a)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & -\frac{\sqrt{210}i}{84} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & -\frac{\sqrt{210}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{42} \\ -\frac{\sqrt{210}}{84} & \frac{\sqrt{210}i}{84} & 0 & 0 & \frac{\sqrt{14}}{14} & -\frac{\sqrt{14}i}{14} & 0 \\ -\frac{\sqrt{210}i}{84} & -\frac{\sqrt{210}}{84} & 0 & 0 & -\frac{\sqrt{14}i}{14} & -\frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{42} & \frac{\sqrt{210}i}{42} & 0 & 0 & 0 \end{bmatrix}$
213	symmetry	$\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_2^{(a)}(E_{2g}^{(b)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & \frac{\sqrt{210}i}{84} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & -\frac{\sqrt{210}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} \\ -\frac{\sqrt{210}}{84} & -\frac{\sqrt{210}i}{84} & 0 & 0 & \frac{\sqrt{14}}{14} & \frac{\sqrt{14}i}{14} & 0 \\ \frac{\sqrt{210}i}{84} & -\frac{\sqrt{210}}{84} & 0 & 0 & \frac{\sqrt{14}i}{14} & -\frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{42} & -\frac{\sqrt{210}i}{42} & 0 & 0 & 0 \end{bmatrix}$
214	symmetry	$\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$
	$\mathbb{Q}_4^{(a)}(A_g)$	$\begin{bmatrix} \frac{3\sqrt{154}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{154}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{154}}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{154}}{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{77} \end{bmatrix}$
215	symmetry	$\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$
	$\mathbb{Q}_4^{(a)}(B_g, 1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}}{22} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 \\ 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 & 0 \\ -\frac{3\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
216	symmetry	$\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_4^{(a)}(B_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}}{22} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{22} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
217	symmetry	$-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$
	$\mathbb{Q}_4^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{1155}}{154} & \frac{\sqrt{1155}i}{154} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{1155}i}{154} & -\frac{\sqrt{1155}}{154} & 0 & 0 & 0 \\ -\frac{\sqrt{1155}}{154} & -\frac{\sqrt{1155}i}{154} & 0 & 0 & \frac{2\sqrt{77}}{77} & -\frac{2\sqrt{77}i}{77} & 0 \\ \frac{\sqrt{1155}i}{154} & -\frac{\sqrt{1155}}{154} & 0 & 0 & \frac{2\sqrt{77}i}{77} & \frac{2\sqrt{77}}{77} & 0 \\ 0 & 0 & \frac{2\sqrt{77}}{77} & \frac{2\sqrt{77}i}{77} & 0 & 0 & \frac{\sqrt{1155}}{154} \\ 0 & 0 & -\frac{2\sqrt{77}i}{77} & \frac{2\sqrt{77}}{77} & 0 & 0 & \frac{\sqrt{1155}i}{154} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{154} & \frac{\sqrt{1155}i}{154} & 0 \end{bmatrix}$
218	symmetry	$-\frac{3\sqrt{5}x^3z}{4} + \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 + \frac{3\sqrt{5}iy^3z}{4} - \sqrt{5}iyz^3$
	$\mathbb{Q}_4^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{1155}}{154} & -\frac{\sqrt{1155}i}{154} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{1155}i}{154} & -\frac{\sqrt{1155}}{154} & 0 & 0 & 0 \\ -\frac{\sqrt{1155}}{154} & \frac{\sqrt{1155}i}{154} & 0 & 0 & \frac{2\sqrt{77}}{77} & \frac{2\sqrt{77}i}{77} & 0 \\ -\frac{\sqrt{1155}i}{154} & -\frac{\sqrt{1155}}{154} & 0 & 0 & -\frac{2\sqrt{77}i}{77} & \frac{2\sqrt{77}}{77} & 0 \\ 0 & 0 & \frac{2\sqrt{77}}{77} & -\frac{2\sqrt{77}i}{77} & 0 & 0 & \frac{\sqrt{1155}}{154} \\ 0 & 0 & \frac{2\sqrt{77}i}{77} & \frac{2\sqrt{77}}{77} & 0 & 0 & -\frac{\sqrt{1155}i}{154} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{154} & -\frac{\sqrt{1155}i}{154} & 0 \end{bmatrix}$
219	symmetry	$\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_4^{(a)}(E_{2g}^{(a)},1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & -\frac{\sqrt{33}i}{22} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{22} & \frac{\sqrt{33}}{22} & 0 \\ 0 & 0 & \frac{\sqrt{55}}{22} & \frac{\sqrt{55}i}{22} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{55}i}{22} & -\frac{\sqrt{55}}{22} & 0 & 0 & 0 \\ -\frac{\sqrt{33}}{22} & -\frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{33}i}{22} & \frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
220	symmetry	$-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$
	$\mathbb{Q}_4^{(a)}(E_{2g}^{(a)},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{3\sqrt{231}}{154} & \frac{3\sqrt{231}i}{154} & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}i}{154} & \frac{3\sqrt{231}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{154} \\ \frac{3\sqrt{231}}{154} & -\frac{3\sqrt{231}i}{154} & 0 & 0 & \frac{\sqrt{385}}{77} & -\frac{\sqrt{385}i}{77} & 0 \\ \frac{3\sqrt{231}i}{154} & \frac{3\sqrt{231}}{154} & 0 & 0 & -\frac{\sqrt{385}i}{77} & -\frac{\sqrt{385}}{77} & 0 \\ 0 & 0 & -\frac{\sqrt{231}}{154} & \frac{\sqrt{231}i}{154} & 0 & 0 & 0 \end{bmatrix}$
221	symmetry	$\frac{\sqrt{70}x^4}{16}-\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}+\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$
	$\mathbb{Q}_4^{(a)}(E_{2g}^{(b)},1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & \frac{\sqrt{33}i}{22} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{22} & \frac{\sqrt{33}}{22} & 0 \\ 0 & 0 & \frac{\sqrt{55}}{22} & -\frac{\sqrt{55}i}{22} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{55}i}{22} & -\frac{\sqrt{55}}{22} & 0 & 0 & 0 \\ -\frac{\sqrt{33}}{22} & \frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{33}i}{22} & \frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
222	symmetry	$-\frac{\sqrt{10}x^4}{8}-\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}-\frac{\sqrt{10}ixy^3}{4}+\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_4^{(a)}(E_{2g}^{(b)},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{3\sqrt{231}}{154} & -\frac{3\sqrt{231}i}{154} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{231}i}{154} & \frac{3\sqrt{231}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{154} \\ \frac{3\sqrt{231}}{154} & \frac{3\sqrt{231}i}{154} & 0 & 0 & \frac{\sqrt{385}}{77} & \frac{\sqrt{385}i}{77} & 0 \\ -\frac{3\sqrt{231}i}{154} & \frac{3\sqrt{231}}{154} & 0 & 0 & \frac{\sqrt{385}i}{77} & -\frac{\sqrt{385}}{77} & 0 \\ 0 & 0 & -\frac{\sqrt{231}}{154} & -\frac{\sqrt{231}i}{154} & 0 & 0 & 0 \end{bmatrix}$
223	symmetry	$-\frac{5x^6}{16}-\frac{15x^4y^2}{16}+\frac{45x^4z^2}{8}-\frac{15x^2y^4}{16}+\frac{45x^2y^2z^2}{4}-\frac{15x^2z^4}{2}-\frac{5y^6}{16}+\frac{45y^4z^2}{8}-\frac{15y^2z^4}{2}+z^6$
	$\mathbb{Q}_6^{(a)}(A_g,1)$	$\begin{bmatrix} -\frac{\sqrt{231}}{462} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{231}}{462} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{231}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{231}}{77} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{10\sqrt{231}}{231} \end{bmatrix}$
224	symmetry	$\frac{\sqrt{462}x^6}{32}-\frac{15\sqrt{462}x^4y^2}{32}+\frac{15\sqrt{462}x^2y^4}{32}-\frac{\sqrt{462}y^6}{32}$
	$\mathbb{Q}_6^{(a)}(A_g,2)$	$\begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
225	symmetry	$\frac{3\sqrt{462}x^5y}{16}-\frac{5\sqrt{462}x^3y^3}{8}+\frac{3\sqrt{462}xy^5}{16}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_6^{(a)}(A_g,3)$	$\begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
226	symmetry	$-\frac{3\sqrt{210}x^5z}{16}+\frac{3\sqrt{210}x^3y^2z}{8}+\frac{\sqrt{210}x^3z^3}{2}+\frac{9\sqrt{210}xy^4z}{16}-\frac{3\sqrt{210}xy^2z^3}{2}$
	$\mathbb{Q}_6^{(a)}(B_g,1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{11} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{11}}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{22} & 0 \\ 0 & 0 & \frac{3\sqrt{11}}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{11}}{22} & 0 & 0 & 0 \\ \frac{\sqrt{11}}{11} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
227	symmetry	$-\frac{9\sqrt{210}x^4yz}{16}-\frac{3\sqrt{210}x^2y^3z}{8}+\frac{3\sqrt{210}x^2yz^3}{2}+\frac{3\sqrt{210}y^5z}{16}-\frac{\sqrt{210}y^3z^3}{2}$
	$\mathbb{Q}_6^{(a)}(B_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{11} \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{11}}{22} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{11}}{22} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{11}}{22} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{11}}{22} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{11}}{11} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
228	symmetry	$\frac{3\sqrt{77}x^5z}{16}-\frac{15\sqrt{77}ix^4yz}{16}-\frac{15\sqrt{77}x^3y^2z}{8}+\frac{15\sqrt{77}ix^2y^3z}{8}+\frac{15\sqrt{77}xy^4z}{16}-\frac{3\sqrt{77}iy^5z}{16}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_6^{(a)}(E_{1g}^{(a)},1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
229	symmetry	$\frac{5\sqrt{42}x^5z}{16}+\frac{5\sqrt{42}ix^4yz}{16}+\frac{5\sqrt{42}x^3y^2z}{8}-\frac{5\sqrt{42}x^3z^3}{4}+\frac{5\sqrt{42}ix^2y^3z}{8}-\frac{5\sqrt{42}ix^2yz^3}{4}+\frac{5\sqrt{42}xy^4z}{16}-\frac{5\sqrt{42}xy^2z^3}{4}+\frac{\sqrt{42}xz^5}{2}+\frac{5\sqrt{42}iy^5z}{16}-\frac{5\sqrt{42}iy^3z^3}{4}+\frac{\sqrt{42}iyz^5}{2}$
	$\mathbb{Q}_6^{(a)}(E_{1g}^{(a)},2)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{33}}{132} & -\frac{\sqrt{33}i}{132} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{33}i}{132} & \frac{\sqrt{33}}{132} & 0 & 0 & 0 \\ \frac{\sqrt{33}}{132} & \frac{\sqrt{33}i}{132} & 0 & 0 & -\frac{\sqrt{55}}{44} & \frac{\sqrt{55}i}{44} & 0 \\ -\frac{\sqrt{33}i}{132} & \frac{\sqrt{33}}{132} & 0 & 0 & -\frac{\sqrt{55}i}{44} & -\frac{\sqrt{55}}{44} & 0 \\ 0 & 0 & -\frac{\sqrt{55}}{44} & -\frac{\sqrt{55}i}{44} & 0 & 0 & \frac{5\sqrt{33}}{66} \\ 0 & 0 & \frac{\sqrt{55}i}{44} & -\frac{\sqrt{55}}{44} & 0 & 0 & \frac{5\sqrt{33}i}{66} \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{66} & \frac{5\sqrt{33}i}{66} & 0 \end{bmatrix}$
230	symmetry	$\frac{3\sqrt{77}x^5z}{16}+\frac{15\sqrt{77}ix^4yz}{16}-\frac{15\sqrt{77}x^3y^2z}{8}-\frac{15\sqrt{77}ix^2y^3z}{8}+\frac{15\sqrt{77}xy^4z}{16}+\frac{3\sqrt{77}iy^5z}{16}$
	$\mathbb{Q}_6^{(a)}(E_{1g}^{(b)},1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
231	symmetry	$\frac{5\sqrt{42}x^5z}{16}-\frac{5\sqrt{42}ix^4yz}{16}+\frac{5\sqrt{42}x^3y^2z}{8}-\frac{5\sqrt{42}x^3z^3}{4}-\frac{5\sqrt{42}ix^2y^3z}{8}+\frac{5\sqrt{42}ix^2yz^3}{4}+\frac{5\sqrt{42}xy^4z}{16}-\frac{5\sqrt{42}xy^2z^3}{4}+\frac{\sqrt{42}xz^5}{2}-\frac{5\sqrt{42}iy^5z}{16}+\frac{5\sqrt{42}iy^3z^3}{4}-\frac{\sqrt{42}iyz^5}{2}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_6^{(a)}(E_{1g}^{(b)},2)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{33}}{132} & \frac{\sqrt{33}i}{132} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{33}i}{132} & \frac{\sqrt{33}}{132} & 0 & 0 & 0 \\ \frac{\sqrt{33}}{132} & -\frac{\sqrt{33}i}{132} & 0 & 0 & -\frac{\sqrt{55}}{44} & -\frac{\sqrt{55}i}{44} & 0 \\ \frac{\sqrt{33}i}{132} & \frac{\sqrt{33}}{132} & 0 & 0 & \frac{\sqrt{55}i}{44} & -\frac{\sqrt{55}}{44} & 0 \\ 0 & 0 & -\frac{\sqrt{55}}{44} & \frac{\sqrt{55}i}{44} & 0 & 0 & \frac{5\sqrt{33}}{66} \\ 0 & 0 & -\frac{\sqrt{55}i}{44} & -\frac{\sqrt{55}}{44} & 0 & 0 & -\frac{5\sqrt{33}i}{66} \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{66} & -\frac{5\sqrt{33}i}{66} & 0 \end{bmatrix}$
232	symmetry	$-\frac{3\sqrt{14}x^6}{32}-\frac{3\sqrt{14}ix^5y}{8}+\frac{15\sqrt{14}x^4y^2}{32}+\frac{15\sqrt{14}x^4z^2}{16}+\frac{15\sqrt{14}ix^3yz^2}{4}+\frac{15\sqrt{14}x^2y^4}{32}-\frac{45\sqrt{14}x^2y^2z^2}{8}+\frac{3\sqrt{14}ixy^5}{8}-\frac{15\sqrt{14}ixy^3z^2}{4}-\frac{3\sqrt{14}y^6}{32}+\frac{15\sqrt{14}y^4z^2}{16}$
	$\mathbb{Q}_6^{(a)}(E_{2g}^{(a)},1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{44} & \frac{\sqrt{110}i}{44} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{44} & -\frac{\sqrt{110}}{44} & 0 \\ 0 & 0 & \frac{\sqrt{66}}{22} & \frac{\sqrt{66}i}{22} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{66}i}{22} & -\frac{\sqrt{66}}{22} & 0 & 0 & 0 \\ \frac{\sqrt{110}}{44} & \frac{\sqrt{110}i}{44} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{110}i}{44} & -\frac{\sqrt{110}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
233	symmetry	$\frac{\sqrt{105}x^6}{32}-\frac{\sqrt{105}ix^5y}{16}+\frac{\sqrt{105}x^4y^2}{32}-\frac{\sqrt{105}x^4z^2}{2}-\frac{\sqrt{105}ix^3y^3}{8}+\sqrt{105}ix^3yz^2-\frac{\sqrt{105}x^2y^4}{32}+\frac{\sqrt{105}x^2z^4}{2}-\frac{\sqrt{105}ixy^5}{16}+\sqrt{105}ixy^3z^2-\sqrt{105}ixyz^4-\frac{\sqrt{105}y^6}{32}+\frac{\sqrt{105}y^4z^2}{2}-\frac{\sqrt{105}y^2z^4}{2}$
	$\mathbb{Q}_6^{(a)}(E_{2g}^{(a)},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{66} & -\frac{\sqrt{33}i}{66} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{66} & -\frac{\sqrt{33}}{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{33}}{33} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{33}i}{33} \\ -\frac{\sqrt{33}}{66} & \frac{\sqrt{33}i}{66} & 0 & 0 & \frac{\sqrt{55}}{22} & -\frac{\sqrt{55}i}{22} & 0 \\ -\frac{\sqrt{33}i}{66} & -\frac{\sqrt{33}}{66} & 0 & 0 & -\frac{\sqrt{55}i}{22} & -\frac{\sqrt{55}}{22} & 0 \\ 0 & 0 & \frac{2\sqrt{33}}{33} & -\frac{2\sqrt{33}i}{33} & 0 & 0 & 0 \end{bmatrix}$
234	symmetry	$-\frac{3\sqrt{14}x^6}{32}+\frac{3\sqrt{14}ix^5y}{8}+\frac{15\sqrt{14}x^4y^2}{32}+\frac{15\sqrt{14}x^4z^2}{16}-\frac{15\sqrt{14}ix^3yz^2}{4}+\frac{15\sqrt{14}x^2y^4}{32}-\frac{45\sqrt{14}x^2y^2z^2}{8}-\frac{3\sqrt{14}ixy^5}{8}+\frac{15\sqrt{14}ixy^3z^2}{4}-\frac{3\sqrt{14}y^6}{32}+\frac{15\sqrt{14}y^4z^2}{16}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_6^{(a)}(E_{2g}^{(b)},1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{44} & -\frac{\sqrt{110}i}{44} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{44} & -\frac{\sqrt{110}}{44} & 0 \\ 0 & 0 & \frac{\sqrt{66}}{22} & -\frac{\sqrt{66}i}{22} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{66}i}{22} & -\frac{\sqrt{66}}{22} & 0 & 0 & 0 \\ \frac{\sqrt{110}}{44} & -\frac{\sqrt{110}i}{44} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{110}i}{44} & -\frac{\sqrt{110}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
235	symmetry	$\frac{\sqrt{105}x^6}{32}+\frac{\sqrt{105}ix^5y}{16}+\frac{\sqrt{105}x^4y^2}{32}-\frac{\sqrt{105}x^4z^2}{2}+\frac{\sqrt{105}ix^3y^3}{8}-\sqrt{105}ix^3yz^2-\frac{\sqrt{105}x^2y^4}{32}+\frac{\sqrt{105}x^2z^4}{2}+\frac{\sqrt{105}ixy^5}{16}-\sqrt{105}ixy^3z^2+\sqrt{105}ixyz^4-\frac{\sqrt{105}y^6}{32}+\frac{\sqrt{105}y^4z^2}{2}-\frac{\sqrt{105}y^2z^4}{2}$
	$\mathbb{Q}_6^{(a)}(E_{2g}^{(b)},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{66} & \frac{\sqrt{33}i}{66} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{66} & -\frac{\sqrt{33}}{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{33}}{33} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{33}i}{33} \\ -\frac{\sqrt{33}}{66} & -\frac{\sqrt{33}i}{66} & 0 & 0 & \frac{\sqrt{55}}{22} & \frac{\sqrt{55}i}{22} & 0 \\ \frac{\sqrt{33}i}{66} & -\frac{\sqrt{33}}{66} & 0 & 0 & \frac{\sqrt{55}i}{22} & -\frac{\sqrt{55}}{22} & 0 \\ 0 & 0 & \frac{2\sqrt{33}}{33} & \frac{2\sqrt{33}i}{33} & 0 & 0 & 0 \end{bmatrix}$
236	symmetry	z
	$\mathbb{M}_1^{(a)}(A_g)$	$\begin{bmatrix} 0 & -\frac{3\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{7} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
237	symmetry	$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{M}_1^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{21}}{28} & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{28} & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ \frac{\sqrt{21}}{28} & \frac{\sqrt{21}i}{28} & 0 & 0 & -\frac{\sqrt{35}}{28} & \frac{\sqrt{35}i}{28} & 0 \\ -\frac{\sqrt{21}i}{28} & \frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{35}i}{28} & -\frac{\sqrt{35}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{35}}{28} & \frac{\sqrt{35}i}{28} & 0 & 0 & -\frac{\sqrt{21}}{14} \\ 0 & 0 & -\frac{\sqrt{35}i}{28} & \frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{21}i}{14} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & \frac{\sqrt{21}i}{14} & 0 \end{bmatrix}$
238	symmetry	$\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$
	$\mathbb{M}_1^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{21}}{28} & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{28} & \frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{28} & \frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{35}}{28} & \frac{\sqrt{35}i}{28} & 0 \\ -\frac{\sqrt{21}i}{28} & -\frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{35}i}{28} & \frac{\sqrt{35}}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{28} & \frac{\sqrt{35}i}{28} & 0 & 0 & \frac{\sqrt{21}}{14} \\ 0 & 0 & -\frac{\sqrt{35}i}{28} & -\frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{21}i}{14} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & \frac{\sqrt{21}i}{14} & 0 \end{bmatrix}$
239	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{M}_3^{(a)}(A_g)$	$\begin{bmatrix} 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
240	symmetry	$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{M}_3^{(a)}(B_g,1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
241	symmetry	$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$
	$\mathbb{M}_3^{(a)}(B_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
242	symmetry	$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$
	$\mathbb{M}_3^{(a)}(E_{1g}^{(a)})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$
243	symmetry	$-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{M}_3^{(a)}(E_{1g}^{(b)})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ \frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{6} & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$
244	symmetry	$\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$
	$\mathbb{M}_3^{(a)}(E_{2g}^{(a)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$
245	symmetry	$\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$
	$\mathbb{M}_3^{(a)}(E_{2g}^{(b)})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$
246	symmetry	$\frac{15x^4z}{8} + \frac{15x^2y^2z}{4} - 5x^2z^3 + \frac{15y^4z}{8} - 5y^2z^3 + z^5$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{M}_5^{(a)}(A_g)$	$\begin{bmatrix} 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{21}i}{21} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{42} & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
247	symmetry	$-\frac{3\sqrt{70}x^4y}{16} - \frac{\sqrt{70}x^2y^3}{8} + \frac{3\sqrt{70}x^2yz^2}{2} + \frac{\sqrt{70}y^5}{16} - \frac{\sqrt{70}y^3z^2}{2}$
	$\mathbb{M}_5^{(a)}(B_g,1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{3} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
248	symmetry	$-\frac{\sqrt{70}x^5}{16} + \frac{\sqrt{70}x^3y^2}{8} + \frac{\sqrt{70}x^3z^2}{2} + \frac{3\sqrt{70}xy^4}{16} - \frac{3\sqrt{70}xy^2z^2}{2}$
	$\mathbb{M}_5^{(a)}(B_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{3} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{3} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
249	symmetry	$\frac{3\sqrt{7}x^5}{16} - \frac{15\sqrt{7}ix^4y}{16} - \frac{15\sqrt{7}x^3y^2}{8} + \frac{15\sqrt{7}ix^2y^3}{8} + \frac{15\sqrt{7}xy^4}{16} - \frac{3\sqrt{7}iy^5}{16}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{M}_5^{(a)}(E_{1g}^{(a)},1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
250	symmetry	$\frac{\sqrt{30}x^5}{16}+\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}+\frac{\sqrt{30}ix^2y^3}{8}-\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}+\frac{\sqrt{30}iy^5}{16}-\frac{3\sqrt{30}iy^3z^2}{4}+\frac{\sqrt{30}iyz^4}{2}$
	$\mathbb{M}_5^{(a)}(E_{1g}^{(a)},2)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{105}}{84} & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{84} & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 \\ \frac{\sqrt{105}}{84} & \frac{\sqrt{105}i}{84} & 0 & 0 & \frac{3\sqrt{7}}{28} & -\frac{3\sqrt{7}i}{28} & 0 \\ -\frac{\sqrt{105}i}{84} & \frac{\sqrt{105}}{84} & 0 & 0 & \frac{3\sqrt{7}i}{28} & \frac{3\sqrt{7}}{28} & 0 \\ 0 & 0 & -\frac{3\sqrt{7}}{28} & -\frac{3\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{105}}{42} \\ 0 & 0 & \frac{3\sqrt{7}i}{28} & -\frac{3\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{105}i}{42} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & \frac{\sqrt{105}i}{42} & 0 \end{bmatrix}$
251	symmetry	$\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$
	$\mathbb{M}_5^{(a)}(E_{1g}^{(b)},1)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
252	symmetry	$\frac{\sqrt{30}x^5}{16}-\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}-\frac{\sqrt{30}ix^2y^3}{8}+\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}-\frac{\sqrt{30}iy^5}{16}+\frac{3\sqrt{30}iy^3z^2}{4}-\frac{\sqrt{30}iyz^4}{2}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{M}_5^{(a)}(E_{1g}^{(b)},2)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{105}}{84} & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{84} & \frac{\sqrt{105}}{84} & 0 & 0 & 0 \\ -\frac{\sqrt{105}}{84} & \frac{\sqrt{105}i}{84} & 0 & 0 & -\frac{3\sqrt{7}}{28} & -\frac{3\sqrt{7}i}{28} & 0 \\ -\frac{\sqrt{105}i}{84} & -\frac{\sqrt{105}}{84} & 0 & 0 & \frac{3\sqrt{7}i}{28} & -\frac{3\sqrt{7}}{28} & 0 \\ 0 & 0 & \frac{3\sqrt{7}}{28} & -\frac{3\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{105}}{42} \\ 0 & 0 & \frac{3\sqrt{7}i}{28} & \frac{3\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{105}i}{42} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & \frac{\sqrt{105}i}{42} & 0 \end{bmatrix}$
253	symmetry	$\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$
	$\mathbb{M}_5^{(a)}(E_{2g}^{(a)},1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
254	symmetry	$-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$
	$\mathbb{M}_5^{(a)}(E_{2g}^{(a)},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & -\frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & -\frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ \frac{\sqrt{6}}{12} & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \end{bmatrix}$
255	symmetry	$\frac{3\sqrt{70}x^4z}{16}-\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}+\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{M}_5^{(a)}(E_{2g}^{(b)},1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
256	symmetry	$-\frac{\sqrt{210}x^4z}{8}-\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}-\frac{\sqrt{210}ixy^3z}{4}+\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$
	$\mathbb{M}_5^{(a)}(E_{2g}^{(b)},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & -\frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ -\frac{\sqrt{6}}{12} & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{12} & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \end{bmatrix}$